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Editors

Nano Letters

American Chemical Society

Dear Editors:

Please consider the enclosed Letter for publication in *Nano Letters*: “Reversible β -Sheet-like Ordering and XPCS-Resolved Dynamic Arrest during Elastin-like Polypeptide Phase Separation.”

Corresponding authors. Qingteng Zhang (qzhang234@anl.gov), Arvind Ramanathan (ramanathana@anl.gov), and Gyorgy Babnigg (gbabnigg@anl.gov).

Significance. Amyloid-like β -sheet packing is usually taken as a route to permanent aggregation; in the sequence-defined elastin-like polypeptide (VPAVG)₃₀ it need not be. Temperature-dependent SAXS/WAXS resolves a reversible transition from a dispersed state at 10 °C to a hierarchical assembly at 30 °C, with an $I(Q) \propto Q^{-3.2}$ low- Q response and peaks at 1.38 and 0.72 Å⁻¹ compatible with β -sheet-like packing. Both signatures vanish on cooling, and seven consecutive 6–34 °C cycles recover the low-temperature profile to within 9%. In a separate isothermal experiment, small-angle X-ray photon correlation spectroscopy (SA-XPCS) at 30 °C resolves a two-step correlation decay whose fast-decay fraction falls from $f = 0.42$ –0.71 to $f = 0.00$ –0.13 over roughly 47 min at every measured Q (76–167 nm). Thermally labile local order therefore coexists with mesoscale dynamics that are progressively arrested, identifying β -sheet-like contacts as candidate switchable physical junctions rather than a commitment to irreversible aggregation.

Novelty. Prior work on gel-forming ELPs established arrested phase separation and network morphology from time-averaged scattering and rheology. This Letter adds the complementary dynamical measurement, asking whether local order stays reversible while the assembly loses mobility. Correlating 100,000 frames per 2 s acquisition at 50 kHz, each at a fresh sample position, separates the fast and slow populations and follows the transfer of weight between them — behavior invisible in the time-averaged SAXS profile, which over the same interval only rises smoothly. To our knowledge this is the first report resolving reversible β -sheet-like ordering and progressive mesoscale arrest in the same sequence-defined protein polymer. We are equally explicit about the limits: the structural and dynamical observations come from separate experiments, so no causal link is claimed; the β -sheet assignment rests on WAXS peak positions, not spectroscopy; the 1.6 s maximum delay bounds a fraction arrested on that scale, not an infinite-time nonergodicity factor; and the isothermal series comes from one loaded sample. All four are stated in the manuscript, and the ordered contacts are called plausible rather than established junctions.

Impact and urgency. The work spans sequence-defined nanomaterials, protein self-assembly, nonequilibrium soft matter, and coherent X-ray characterization, and gives a nanoscale structure–dynamics framework for responsive protein materials whose β -sheet-like order switches thermally without permanent aggregation. Because the behavior is sequence-encoded, the strategy transfers to other programmable protein polymers. High-speed detectors and upgraded storage rings have only recently made XPCS at these count rates practical on weakly scattering biological samples, so a concise, carefully bounded demonstration is timely. The reduced data and all analysis and figure-generation code are already public on GitHub (qzhang234/PubCode-2021-IDPP).

The manuscript is original, is not under consideration elsewhere, and has been approved by all authors. The authors declare no competing financial interest; AI-tool use is disclosed in the Acknowledgments. Thank you for your consideration.

Sincerely,

Qingteng Zhang, Ph.D., on behalf of all authors

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Suggested Reviewers

The following researchers have relevant expertise in elastin-like polypeptides, protein-based materials, arrested phase separation, and X-ray photon correlation spectroscopy.

Bradley D. Olsen

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Expertise: elastin-like polypeptide gels, associative protein polymers, and arrested phase separation.

Ashutosh Chilkoti

Alan L. Kaganov Distinguished Professor of Biomedical Engineering, Duke University
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Expertise: recombinant elastin-like polypeptides, thermoresponsive protein materials, and hierarchical self-assembly.

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Expertise: protein phase separation, SAXS, and XPCS of protein-rich systems.

Christian Gutt

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Expertise: XPCS, nonequilibrium dynamics, and dense protein systems.

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Expertise: colloidal gelation, arrested phase separation, and network-forming fluids.